Republic of Iraq
Ministry of Higher Education and Scientific Research
University of Baghdad
College of Medicine

Health Anxiety, AI Chatbot Use and Cyberchondria Among Adults in Baghdad: An Online Cross-Sectional Survey

A graduation research project submitted to the College of Medicine, University of Baghdad, in partial fulfilment of the requirements for the degree of [Degree]

A research project presented by

Abdulrahman Jamal Ghazi

Mustafa Wamedh Mustafa

Yaseen Omar Lutfi

Supervised by

Assist. Prof. Dr. Thikra Hasan Mathkor

[Academic year]

Acknowledgment

[To be completed by the authors.]

We thank all the participants who gave their time to complete this survey, and [names of supervisor, department and any others to be acknowledged].

Table of Contents

Summary / Abstract.....viii
1. Introduction.....1
1.1 Background.....1
1.2 Health anxiety.....1
1.3 Cyberchondria.....2
1.4 Health anxiety and cyberchondria.....2
1.5 From search engines to AI chatbots.....2
1.6 Searching and the doctor–patient relationship.....3
1.7 Rationale.....3
1.8 Aim, objectives and hypotheses.....3
2. Method.....5
2.1 Design, setting and participants.....5
2.2 Sample size.....5
2.3 Questionnaire.....5
2.4 Definitions.....6
2.5 Statistical analysis.....6
2.6 Ethical considerations.....7
3. Results.....7
3.1 Participants and online health-information seeking.....7
3.2 Levels of health anxiety and cyberchondria.....10
3.3 Association between health anxiety and cyberchondria.....11
3.4 Factors associated with health anxiety and cyberchondria.....12
3.5 Independent predictors.....16
3.6 Behaviour and pathways.....18
4. Discussion.....20
4.1 Health anxiety and cyberchondria.....20
4.2 Do health anxiety and cyberchondria fuel each other?.....21
4.3 AI chatbots.....21
4.4 The consultation and medical students.....22
4.5 Strengths and limitations.....22
5. Conclusions and Recommendations.....23
5.1 Conclusions.....23

5.2 Recommendations.....24
6. References.....24
Appendix. Study questionnaire.....28

List of Abbreviations

AI	Artificial intelligence
AOR	Adjusted odds ratio
AUC	Area under the receiver operating characteristic curve
CFA	Confirmatory factor analysis
CFI	Comparative fit index
CI	Confidence interval
CSS-12	Cyberchondria Severity Scale, 12-item short form
DWLS	Diagonally weighted least squares
ER	Emergency room
FDR	False discovery rate
GP	General practitioner
HA	Health anxiety
HC3	Heteroskedasticity-consistent standard errors, type 3
IQR	Interquartile range
KMO	Kaiser–Meyer–Olkin measure of sampling adequacy
LLM	Large language model
ML	Maximum likelihood
OLS	Ordinary least squares
OR	Odds ratio
RMSEA	Root mean square error of approximation
SD	Standard deviation
SHAI-14	Short Health Anxiety Inventory, 14-item version
STROBE	Strengthening the Reporting of Observational Studies in Epidemiology
TLI	Tucker–Lewis index
VIF	Variance inflation factor

List of Tables

Table 1. Socio-demographic characteristics of the participants (n = 300).....8

Table 2. Online health information seeking, pre-consultation behaviour and relationship with physicians (n = 300).....9

Table 3. Descriptive statistics, reliability and language equivalence of the SHAI-14 and CSS-12 (n = 300).....11

Table 4. Prevalence of elevated health anxiety at different SHAI-14 cut-offs (n = 300).....11

Table 5. Correlation between health anxiety (SHAI-14) and cyberchondria (CSS-12) and its dimensions (n = 300).....12

Table 6. Health anxiety (SHAI-14) and cyberchondria (CSS-12) scores by participant characteristics (n = 300).....14

Table 7. Variables significantly associated ($p < 0.05$) with elevated health anxiety and with high cyberchondria.....15

Table 8. Hierarchical robust linear regression of cyberchondria (CSS-12 total) – final model (n = 300).....17

Table 9. Multivariable logistic regression of elevated health anxiety (SHAI ≥ 18; 57 events, n = 300).....17

Table 10. Adjusted associations of health anxiety and cyberchondria with pre-consultation behaviours and the doctor–patient relationship.....18

Table 11. Exploratory mediation models decomposing the association between health anxiety (HA) and the dimensions of cyberchondria (standardised coefficients, n = 300).....19

List of Figures

Figure 1. Conceptual framework of the study: health anxiety is associated with cyberchondria (H1) independently of covariates (H2), partly through excessive searching and post-search distress (H3), and cyberchondria with pre-consultation behaviour (H4).....4

Figure 2. Tools used and motivations for searching for health information online (n = 300).....10

Figure 3. Behaviour after searching, disclosure of searches to physicians and perceived effect on the doctor–patient relationship (n = 300).....10

Figure 4. Health anxiety (SHAI-14) against cyberchondria (CSS-12) with the least-squares line and 95% confidence band (n = 300).....12

Figure 5. Cyberchondria (CSS-12) by online tool profile.....15

Figure 6. Adjusted odds ratios for elevated health anxiety (SHAI ≥ 18); orange: p < 0.05.....18

Figure 7. Mediation models: (A) excessive searching between health anxiety and post-search distress; (B) post-search distress between health anxiety and reassurance seeking.....20

Figure 8. Cyberchondria dimensions by health-anxiety status (mean ± 95% CI).. 20

Summary / Abstract

Background: Many adults look up symptoms online before seeing a doctor, increasingly with artificial-intelligence (AI) chatbots. Health anxiety and cyberchondria (excessive, distressing online health searching) are thought to feed each other, but evidence from Iraq, and on AI chatbots, is scarce.

Objectives: To measure the association between health anxiety and cyberchondria among adults in Baghdad, and to describe online and AI health-information seeking and behaviour before consulting a doctor.

Methods: Cross-sectional online survey of 300 adults in Baghdad (July–September 2026) using the Short Health Anxiety Inventory (SHAI-14) and the Cyberchondria Severity Scale (CSS-12), analysed with FDR-corrected group comparisons, robust regression and bootstrapped mediation.

Results: Participants were mostly young (62.3% aged 20–29). AI chatbots were ticked by 78.0% (88.7% including those who named a chatbot), more than search engines (56.3%). Health anxiety correlated with cyberchondria ($r = 0.47$, 95% CI 0.38–0.55) and remained its strongest predictor after adjustment ($\beta = 0.48$), most strongly with excessive searching and post-search distress. Elevated health anxiety (SHAI-14 ≥ 18, a screening threshold, not a diagnosis) was found in 19.0%. After searching, 34.7% delayed or skipped care because they felt reassured and 31.7% never told their doctor, citing fear of judgement. The association between AI use and cyberchondria was not robust to how AI use was defined.

Conclusion: In this young, health-oriented convenience sample, health anxiety and cyberchondria were linked and AI chatbots were a major source of health information, without reliable evidence that they add to cyberchondria beyond health anxiety. Clinicians should ask patients, without judgement, what they have searched or asked an AI.

Keywords: cyberchondria; health anxiety; artificial intelligence; chatbots; Iraq.

1. Introduction

1.1 Background

Many people now look a symptom up online before they see a doctor. Often it goes into a search engine, or is described to an AI chatbot, well before a physician hears about it. The information is immediate, free and available at any hour, and it can help people decide whether a symptom needs medical attention. For a sizeable minority, however, searching makes the worry worse, a pattern known as cyberchondria [1,2]. About four in five Iraqis were online by 2025 [3], yet we found no study from Baghdad on how this habit interacts with health anxiety or affects the consultation that follows.

1.2 Health anxiety

Health anxiety is a persistent preoccupation with having, or getting, a serious illness, kept going by misreading normal bodily sensations. It runs on a continuum from passing worry to illness anxiety disorder and somatic symptom disorder in the DSM-5 [4]. In the cognitive-behavioural model, sensations are read as signs of serious disease, and the resulting anxiety is maintained by safety behaviours such as body checking and reassurance seeking, which relieve it briefly but stop the person learning that the feared outcome does not happen [5].

Health anxiety is common and costly: its lifetime prevalence is about 6% in the general population and up to 20% in hospital outpatients, and it generates unnecessary medical contacts [6]. In Iraq, 49.5% of 1591 respondents to an online survey during the COVID-19 restrictions screened positive for health anxiety on the Health Anxiety Questionnaire [7].

1.3 Cyberchondria

White and Horvitz showed in large-scale search logs that searches for common, harmless symptoms often escalated towards rare and serious diagnoses [1]. Starcevic and Berle defined cyberchondria as excessive or repeated online health searching that increases, rather than relieves, distress [2]; searching itself is not pathological, but becomes cyberchondria when it is excessive, compulsive and distressing. The 33-item Cyberchondria Severity Scale has a 12-item short form (CSS-12) that keeps four subscales (excessiveness, compulsion, distress and reassurance seeking) [8], and an Arabic version has shown acceptable structure in Lebanon [9].

1.4 Health anxiety and cyberchondria

The two are closely related but not identical: beyond its overlap with health anxiety, cyberchondria is linked with more functional impairment and healthcare use [10], and in a randomised experiment searching the causes of provoked bodily symptoms raised health anxiety and the wish to see a physician [11]. Two meta-analyses found that health anxiety correlates strongly with cyberchondria ($r = 0.62$ and 0.63), with subscale correlations from 0.24 to 0.66 [12,13]. Whether the two fuel each other is unsettled. A four-wave Dutch panel of 5322 adults found early evidence of a two-way relationship, though only in people with non-clinical health anxiety [14].

1.5 From search engines to AI chatbots

Most of this evidence predates large language model chatbots such as ChatGPT and Gemini, which answer in fluent, personal language, can be questioned again and again, and can seem more empathetic than physicians [15]. In a large US survey about one in five respondents had used ChatGPT for health information, and about a third of them changed medications or asked for referrals based on its answers [16]. It remains unclear whether

chatbots relieve worry with calm answers or feed it by always being available for another round of reassurance.

1.6 Searching and the doctor–patient relationship

Online information can help the relationship when patients discuss it with their physician, and strain it when they hide it [17]. Reassurance seeking may lead to extra consultations, while false reassurance may delay care. In the Arab region, cyberchondria was common in Jeddah [18] and linked with health anxiety in Saudi primary care [19], and across 13 Arabic-speaking countries trust in online information predicted delayed consultation [20]. No comparable study exists from Iraq.

1.7 Rationale

This study addresses three gaps: it gives, to our knowledge, the first estimate of the association between health anxiety and cyberchondria (CSS-12) in Baghdad; it examines AI chatbots separately from search engines; and it links both constructs to what clinicians see before and during a consultation.

1.8 Aim, objectives and hypotheses

Aim: To study the relationship between health anxiety and cyberchondria among adults in Baghdad who search for health information online, with a focus on AI chatbots and pre-consultation behaviour.

Objectives:

1. To describe where adults in Baghdad look for health information (online and through AI) and what they do before seeing a doctor.
2. To estimate levels of health anxiety and cyberchondria and the strength of their association, independently of socio-demographic and digital-use factors (H1, H2).

3. To examine statistical pathways consistent with a mutually reinforcing cycle (H3).
4. To examine how health anxiety and cyberchondria relate to pre-consultation behaviour, disclosure of searches and trust in physicians (H4).

Hypotheses: (H1) People with more health anxiety have more cyberchondria, especially on the distress and excessiveness dimensions. (H2) This holds after adjustment for socio-demographic and digital-use factors. (H3) Part of the link between health anxiety and post-search distress runs, statistically, through excessive searching. (H4) People with more cyberchondria make more urgent visits and multiple consultations and report a more strained doctor–patient relationship. H1 and H2 were primary and H3 and H4 secondary; all other analyses were exploratory (Figure 1).

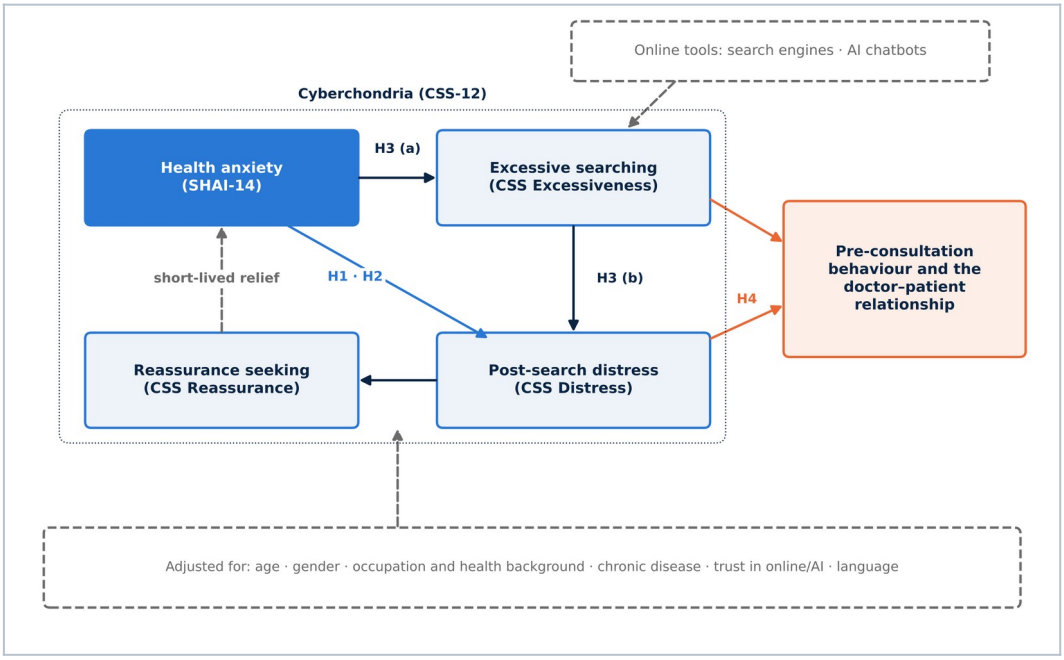


Figure 1. *Conceptual framework of the study: health anxiety is associated with cyberchondria (H1) independently of covariates (H2), partly through excessive searching and post-search distress (H3), and cyberchondria with pre-consultation behaviour (H4).*

2. Method

2.1 Design, setting and participants

This analytical cross-sectional study follows the STROBE [21] and CHERRIES [22] reporting guidelines. Data were collected online in Baghdad between 17 July 2026 and 11 September 2026 by convenience and snowball sampling through Baghdad-based social-media and messaging groups. Adults aged 18 or over who could read Arabic or English and gave informed consent were eligible. All completed submissions were analysed; no duplicate response patterns were found [Authors to state whether Google Forms limited submissions to one per account]. Three responses annotated by the team during data collection [authors to state the reason] were kept and tested in a sensitivity analysis.

2.2 Sample size

Using Fisher's z with two-sided $\alpha = 0.05$ and 80% power, 194 participants are needed to detect a correlation of $r = 0.20$ [23]; meta-analytic estimates are far larger ($r \approx 0.62$), so the target is conservative. For proportions, 300 participants give a margin of error of ±5.7%.

2.3 Questionnaire

A bilingual (Arabic–English) Google Form had five parts: (1) consent and socio-demographic data; (2) digital health-information seeking (tools, main AI chatbot, motives and trust in online/AI information); (3) the 14-item Short Health Anxiety Inventory (SHAI-14; items scored 0–3, total 0–42) [24], which has good reliability and validity [25,26]; (4) the CSS-12 (items scored 1–5, total 12–60; four 3-item subscales) [8]; and (5) behaviour after searching, disclosure of searches to the physician and the effect on the relationship with physicians. The Arabic wording was written by the team

without formal cross-cultural adaptation [27]; published Arabic versions [9,28] were not used. The response options of one SHAI item (item 2) were displayed in a different order from the original and were scored by meaning.

2.4 Definitions

- **Elevated health anxiety:** SHAI-14 ≥ 18, the cut-off that best separated people with health anxiety from healthy controls in a Swedish diagnostic study [26]; it is a screening threshold, not a diagnosis, so prevalence at 15, 22 and 29 is also reported.
- **High cyberchondria:** CSS-12 > 33 (upper tertile), as the CSS-12 has no validated cut-off.
- **AI-chatbot user:** ticked "AI chatbots" or named a chatbot; because 32 respondents named one without ticking it, all AI analyses were repeated with the strict (ticked-only) definition.
- **Health-related background:** medical students, other health-science students and healthcare workers.

2.5 Statistical analysis

Analyses were scripted in Python 3 (two-sided $\alpha = 0.05$). Reliability was assessed with Cronbach's α and McDonald's ω [29], structure with confirmatory factor analysis (maximum likelihood), and equivalence of Arabic and English responders with multi-group measurement invariance [30] and item-level differential item functioning [31]. Associations were tested with Pearson and Spearman correlations (bootstrap CIs) and partial correlations; dependent correlations were compared with Williams' T2 test [32]. Scores were compared between groups with Mann–Whitney U or Kruskal–Wallis tests and categorical variables with χ^2 or Fisher's exact tests. Because many tests were run, p-values within each family were adjusted by

the Benjamini–Hochberg false discovery rate (q-values) [33], and associations that did not survive correction are called exploratory.

Multivariable analyses comprised hierarchical robust (HC3) linear regression of CSS on socio-demographics, digital use and SHAI; logistic regression of elevated health anxiety (AUC corrected for optimism by bootstrap) and of each behaviour; and exploratory mediation models with 5000 bootstrap resamples [34], plus moderation by interaction terms. Sensitivity analyses excluded annotated and straight-line responses, removed SHAI item 2 and compared early with late responders.

2.6 Ethical considerations

Participation was voluntary and anonymous; electronic informed consent (with confirmation of age ≥ 18) was required before any question, and no names or contact details were collected. [Ethical approval: committee name, approval number and date to be inserted.]

3. Results

3.1 Participants and online health-information seeking

All 300 participants completed the questionnaire with no missing data. The sample was young (62.3% aged 20–29), 56.3% were female, 44.0% were medical students and 63.3% had a health-related background (Table 1). AI chatbots were the most used tool (78.0% ticked; 88.7% including those who named one), ahead of search engines (56.3%); 41.0% used AI without any search engine and ChatGPT was used by 59.3%. AI use fell from 96% at 18–19 years to 72% at 40 or older. The leading motives were convenience, deciding whether a doctor was needed and instant reassurance (Table 2, Figure 2).

Table 1. Socio-demographic characteristics of the participants (n = 300)

Characteristic	n	%
Age group (years)		
18-19	56	18.7
20-29	187	62.3
30-39	21	7.0
40-49	25	8.3
50-59	9	3.0
60+	2	0.7
Gender		
Female	169	56.3
Male	131	43.7
Highest education		
Currently a student	189	63.0
High school or below	18	6.0
Diploma/Vocational	7	2.3
Bachelor	65	21.7
Postgraduate	21	7.0
Study/work		
Medical student	132	44.0
Other health-science student	43	14.3
Healthcare worker	15	5.0
Non-health student	46	15.3
Non-health employee/other	64	21.3
Health-related background (1=yes)		
Yes	190	63.3
No	110	36.7
Diagnosed chronic condition (1=yes)		
Yes	65	21.7
No	235	78.3
Language relied on		
Arabic	212	70.7
English	88	29.3

Table 2. Online health information seeking, pre-consultation behaviour and relationship with physicians (n = 300)

Variable	n	%
Tools used (multiple)		
AI chatbots*	266	88.7
Search engines	169	56.3
Dedicated medical platforms	55	18.3
Social media	32	10.7
Main AI chatbot		
ChatGPT	178	59.3
Gemini	71	23.7
Claude	12	4.0
Other AI	3	1.0
None reported	36	12.0
Tool profile		
AI + search engine	143	47.7
AI without search engine	123	41.0
No AI (search/other only)	34	11.3
Motivation (multiple)		
Convenience/accessibility	240	80.0
Check if doctor visit is needed	173	57.7
Instant reassurance	153	51.0
Self-diagnose / double-check prescription	69	23.0
Cost/logistic barriers	54	18.0
Trust in online/AI medical data (1-5)		
1 – Do not trust at all	20	6.7
2 – Slightly	83	27.7
3 – Moderately	170	56.7
4 – Very much	25	8.3
5 – Completely	2	0.7
Behaviour after searching (multiple)		
Delay/skip care (reassured)	104	34.7
Ignore symptom	90	30.0
Visit multiple doctors	76	25.3
Urgent visit/ER/fast diagnosis	66	22.0
Self-medication	64	21.3
Delay care (terrified/overwhelmed)	26	8.7
Disclose search to physician		
Always	81	27.0
Sometimes	124	41.3
Never (fear of judgement)	95	31.7
Effect on doctor relationship		
No negative impact	196	65.3
Somewhat complicated	66	22.0
Significantly reduced trust	38	12.7

Note. Tools, motivations and behaviours allowed multiple answers, so percentages do not sum to 100. AI users include 32 respondents who named a chatbot without ticking the AI option.

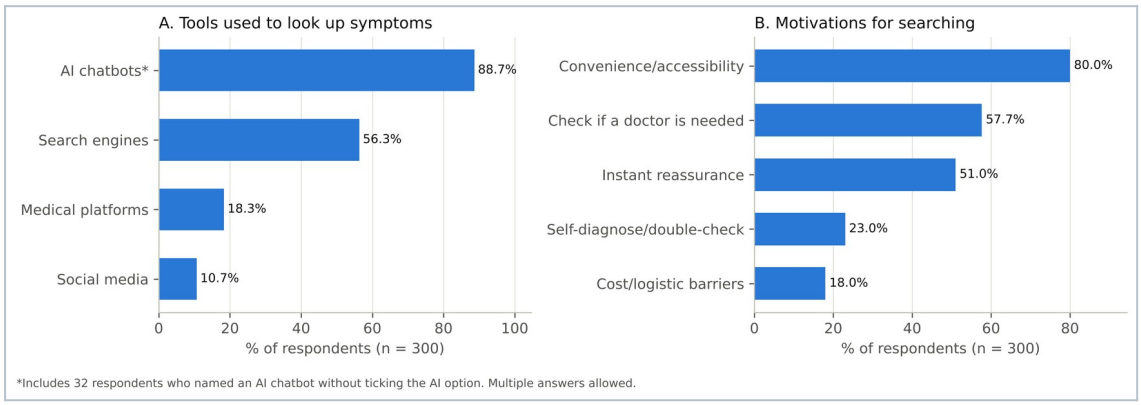


Figure 2. *Tools used and motivations for searching for health information online (n = 300).*

After searching, 34.7% delayed or skipped care because the information was reassuring, 30.0% ignored the symptom, 25.3% consulted several doctors and 22.0% sought urgent care. Only 27.0% always told their physician about their searches, whereas 31.7% never did, citing fear of judgement, and 34.7% said searching had complicated their relationship with, or trust in, their doctor (Figure 3).

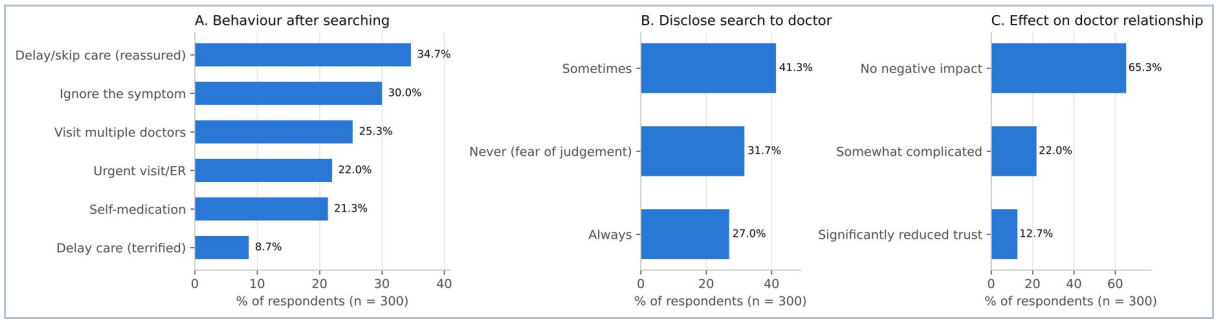


Figure 3. *Behaviour after searching, disclosure of searches to physicians and perceived effect on the doctor–patient relationship (n = 300).*

3.2 Levels of health anxiety and cyberchondria

Scale scores and reliability are shown in Table 3. The mean SHAI-14 score was 12.35 ± 5.83; 57 (19.0%) (95% CI 15.0–23.8%) had elevated health anxiety, an estimate that ranged from 32.7% at a cut-off of 15 to 0.7% at 29 (Table 4). The mean CSS-12 score was 28.86 ± 8.36; participants searched frequently (excessiveness was the highest dimension) but searching rarely interfered with daily life (compulsion was the lowest). Both scales were

reliable (α = 0.81 and 0.86), the four-factor CSS-12 structure fitted acceptably (CFI = 0.954, RMSEA = 0.070), and the CSS-12 was fully invariant across Arabic and English responders; the SHAI-14 reached metric but not scalar invariance from a weak baseline, so its language comparisons need caution.

Table 3. Descriptive statistics, reliability and language equivalence of the SHAI-14 and CSS-12 (n = 300)

Scale	Mean ± SD	Median (IQR)	Range	α (95% CI)	ω
SHAI-14 total (0–42)	12.35 ± 5.83	12 (8–16)	1–30	0.81 (0.78–0.84)	0.82
CSS-12 total (12–60)	28.86 ± 8.36	29 (23–34)	12–57	0.86 (0.84–0.89)	0.87
Excessiveness (3–15)	9.66 ± 2.94	10 (8–12)	3–15	0.77 (0.72–0.81)	0.77
Compulsion (3–15)	5.00 ± 2.58	4 (3–6)	3–15	0.88 (0.86–0.90)	0.89
Distress (3–15)	7.24 ± 2.74	7 (5–9)	3–15	0.76 (0.71–0.81)	0.77
Reassurance (3–15)	6.96 ± 2.81	7 (5–9)	3–15	0.76 (0.71–0.81)	0.77

Note. Language equivalence (SHAI-14 means are not strictly comparable across languages because its scalar invariance was not supported): SHAI mean 12.40±5.73 (Arabic) vs 12.23±6.08 (English), adjusted p = 0.52; CSS mean 28.96±8.21 vs 28.62±8.77, adjusted p = 0.89. CFA fit, ML (primary): CSS-12 four factors CFI 0.954, TLI 0.937, RMSEA 0.070, SRMR 0.053; SHAI-14 one factor CFI 0.830, RMSEA 0.079, SRMR 0.064. DWLS on Pearson covariances (comparison only, not comparable with ML cut-offs): CSS-12 CFI 1.000, RMSEA 0.003; SHAI-14 CFI 0.977, RMSEA 0.038. Measurement invariance (multi-group CFA): CSS-12 scalar invariant; SHAI-14 metric but not scalar invariant; no item DIF after FDR correction.

Table 4. Prevalence of elevated health anxiety at different SHAI-14 cut-offs (n = 300)

SHAI-14 score	Basis of the cut-off	n	%	95% CI
≥ 15	Lower threshold (sensitivity)	98	32.7	27.6–38.2
≥ 18	**Optimal versus healthy controls (primary definition)**	**57**	**19.0**	**15.0–23.8**
≥ 22	Optimal in a psychiatric setting	22	7.3	4.9–10.9
≥ 29	Optimal in a low-prevalence setting	2	0.7	0.2–2.4

Note. Cut-offs of 18, 22 and 29 from the Swedish diagnostic-accuracy study of the SHAI-14 [26]; ≥ 18 is used throughout this thesis. 95% CI: Wilson score interval. An elevated score is a screening result, not a diagnosis.

3.3 Association between health anxiety and cyberchondria

Health anxiety and cyberchondria were moderately correlated (r = 0.47, 95% CI 0.38–0.55; Spearman ρ = 0.47; p < 0.001), and the correlation was unchanged after adjustment for age, gender, health background, chronic disease, trust and AI use (partial r = 0.48) (Table 5, Figure 4). Correlations

with excessiveness (r = 0.46) and distress (r = 0.45) were significantly stronger than with compulsion and reassurance seeking (Williams' T2, p < 0.001). Item by item, cyberchondria followed fear of serious illness and the interpretation of bodily sensations, not mere awareness of sensations; 111 of 168 item pairs remained significant after FDR correction.

Table 5. Correlation between health anxiety (SHAI-14) and cyberchondria (CSS-12) and its dimensions (n = 300)

Cyberchondria measure	Pearson r (95% CI)	Spearman ρ	p	Adjusted β
CSS-12 total	0.47 (0.38–0.55)	0.47	<0.001	0.48
Excessiveness	0.46 (0.37–0.55)	0.45	<0.001	0.44
Compulsion	0.25 (0.14–0.36)	0.32	<0.001	0.29
Distress	0.45 (0.36–0.54)	0.44	<0.001	0.44
Reassurance	0.24 (0.13–0.35)	0.23	<0.001	0.25

Note. Adjusted β: standardised coefficient of SHAI from robust OLS models adjusted for age, gender, occupation, chronic disease, language, tools, trust and motives. Excessiveness and Distress correlations were each significantly stronger than Compulsion and Reassurance (Williams' T2, p < 0.001); they did not differ from each other (p = 0.81).

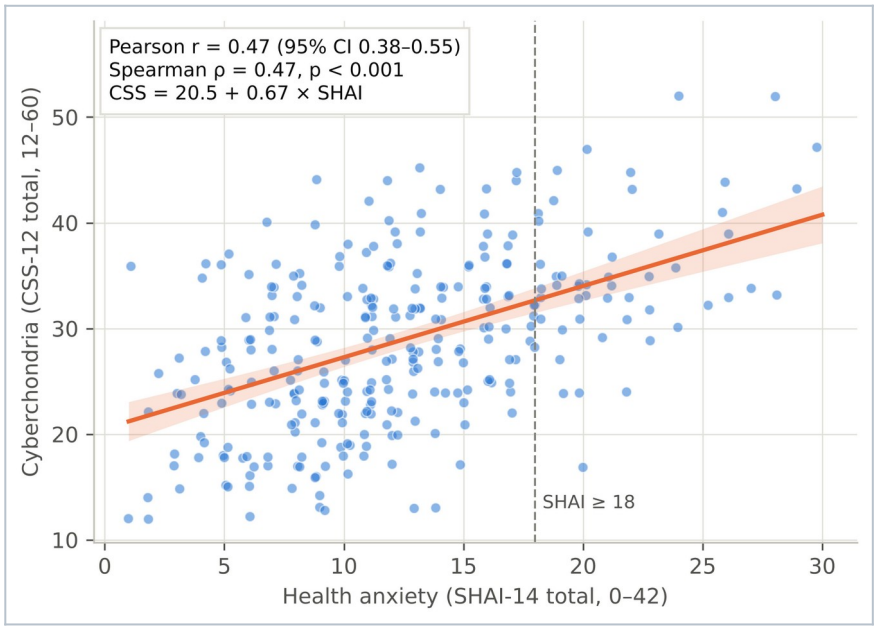


Figure 4. *Health anxiety (SHAI-14) against cyberchondria (CSS-12) with the least-squares line and 95% confidence band (n = 300).*

3.4 Factors associated with health anxiety and cyberchondria

Across 162 group comparisons (Table 6), health anxiety was higher with a chronic condition (q = 0.028) and, most clearly, in those searching for instant reassurance (q = 0.002); differences by age, education and occupation

were borderline and exploratory, and gender made no difference. Under the inclusive definition, AI users had higher cyberchondria ($q = 0.005$), highest in those combining AI with search engines (Figure 5). Participants with elevated health anxiety more often had high cyberchondria (52.6% vs 23.0%; OR = 3.71) (Table 7).

Table 6. Health anxiety (SHAI-14) and cyberchondria (CSS-12) scores by participant characteristics (n = 300)

Variable	n	SHAI mean ± SD	p (q)	CSS mean ± SD	p (q)
Age group			**0.008 (0.051)**		**0.288 (0.505)**
18-19	56	14.0 ± 5.5		28.6 ± 8.1	
20-29	187	12.4 ± 5.8		28.4 ± 8.0	
30-39	21	10.2 ± 6.4		32.5 ± 11.3	
≥40	36	11.0 ± 5.7		29.7 ± 8.2	
Gender			**0.378 (0.589)**		**0.859 (0.918)**
Female	169	12.7 ± 6.0		29.0 ± 8.7	
Male	131	11.9 ± 5.5		28.6 ± 8.0	
Study/work			**0.005 (0.039)**		**0.820 (0.918)**
Medical student	132	13.3 ± 5.8		28.9 ± 8.1	
Other health-science student	43	12.2 ± 5.5		29.2 ± 9.1	
Healthcare worker	15	10.4 ± 5.1		29.5 ± 7.4	
Non-health student	46	13.0 ± 6.0		27.6 ± 7.8	
Non-health employee/other	64	10.5 ± 5.7		29.4 ± 9.1	
Chronic condition			**0.004 (0.028)**		**0.435 (0.623)**
Yes	65	14.2 ± 6.0		29.9 ± 8.9	
No	235	11.9 ± 5.7		28.6 ± 8.2	
Uses AI chatbots			**0.008 (0.051)**		**<0.001 (0.005)**
Yes	266	12.6 ± 5.7		29.5 ± 8.2	
No	34	10.5 ± 6.7		24.0 ± 8.1	
Uses search engines			**0.111 (0.316)**		**0.113 (0.316)**
Yes	169	12.8 ± 5.9		29.4 ± 7.9	
No	131	11.8 ± 5.7		28.2 ± 8.9	
Uses social media			**0.001 (0.016)**		**0.021 (0.093)**
Yes	32	15.8 ± 6.9		32.5 ± 9.1	
No	268	11.9 ± 5.6		28.4 ± 8.2	
Tool profile			**0.006 (0.039)**		**<0.001 (0.005)**
AI + search engine	143	13.1 ± 5.6		30.1 ± 7.6	
AI without search engine	123	12.0 ± 5.8		28.7 ± 8.8	
No AI (search/other only)	34	10.5 ± 6.7		24.0 ± 8.1	
Motive: reassurance			**<0.001 (0.002)**		**0.085 (0.280)**
Yes	153	13.8 ± 5.7		29.6 ± 7.6	
No	147	10.9 ± 5.7		28.1 ± 9.1	
Motive: self-diagnosis			**0.217 (0.435)**		**0.079 (0.266)**
Yes	69	13.1 ± 6.2		30.4 ± 8.1	
No	231	12.1 ± 5.7		28.4 ± 8.4	
Trust in online/AI			**0.052 (0.190)**		**0.110 (0.316)**
1	20	10.1 ± 6.4		24.8 ± 7.8	
2	83	13.2 ± 6.1		28.9 ± 8.3	
3	170	11.9 ± 5.6		28.9 ± 8.4	
4	25	14.7 ± 5.2		31.8 ± 7.1	
5	2	13.0 ± 1.4		25.5 ± 17.7	
Language			**0.857 (0.918)**		**0.654 (0.802)**
Arabic	212	12.4 ± 5.7		29.0 ± 8.2	
English	88	12.2 ± 6.1		28.6 ± 8.8	

Note. SHAI-14 comparisons by language should be read cautiously because SHAI-14 scalar invariance was not supported. p: Mann–Whitney U (two groups) or Kruskal–Wallis H (three or more groups); q: Benjamini–Hochberg false-discovery-rate adjusted p across all 162 group comparisons. Effect sizes, medians (IQR), Welch tests and post-hoc comparisons are available from the authors.

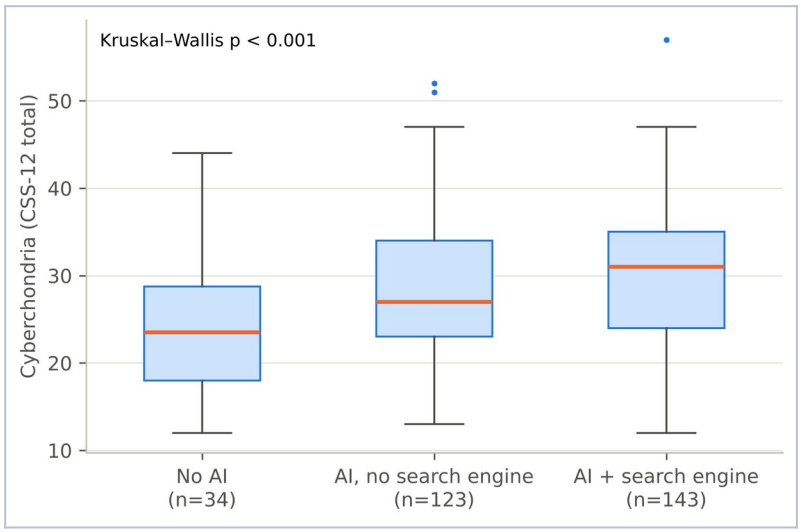


Figure 5. *Cyberchondria (CSS-12) by online tool profile.*

Table 7. Variables significantly associated ($p < 0.05$) with elevated health anxiety and with high cyberchondria

Variable	n/N (%) with outcome, by group	Cramér's V	p	q (FDR)
Elevated health anxiety (SHAI ≥ 18)				
Motive: reassurance	No: 15/147 (10.2%); Yes: 42/153 (27.5%)	0.22	<0.001	0.004
Uses social media	No: 44/268 (16.4%); Yes: 13/32 (40.6%)	0.19	0.001	0.015
Delay care (terrified)	No: 47/274 (17.2%); Yes: 10/26 (38.5%)	0.15	0.016	0.107
Motive: convenience	No: 17/60 (28.3%); Yes: 40/240 (16.7%)	0.12	0.039	0.180
High cyberchondria (CSS > 33)				
Elevated health anxiety	No: 56/243 (23.0%); Yes: 30/57 (52.6%)	0.26	<0.001	0.001
Disclose to physician	Always: 33/81 (40.7%); Never : 15/95 (15.8%); Sometimes: 38/124 (30.6%)	0.21	0.001	0.015
Ignore symptom	No: 70/210 (33.3%); Yes: 16/90 (17.8%)	0.16	0.006	0.060
Uses AI chatbots	No: 3/34 (8.8%); Yes: 83/266 (31.2%)	0.16	0.007	0.060
Tool profile	AI + search engine: 49/143 (34.3%); AI without search engine: 34/123 (27.6%); No AI (search/other only): 3/34 (8.8%)	0.17	0.012	0.096
Effect on doctor relationship	No negative impact: 46/196 (23.5%); Significantly reduced trust: 14/38 (36.8%); Somewhat complicated: 26/66 (39.4%)	0.16	0.023	0.141
Main AI chatbot	ChatGPT: 61/178 (34.3%); Claude: 4/12 (33.3%); Gemini: 17/71 (23.9%); None reported: 4/36 (11.1%); Other AI: 0/3 (0.0%)	0.18	0.034	0.180
Delay care (terrified)	No: 74/274 (27.0%); Yes: 12/26 (46.2%)	0.12	0.039	0.180

Note. Pearson χ^2 or Fisher's exact test (Monte-Carlo exact p where expected counts were < 5). Associations with q ≥ 0.05 are exploratory. Non-significant associations are available from the authors.

3.5 Independent predictors

In hierarchical regression (Table 8), socio-demographics explained 3% of the variance in cyberchondria, digital use raised this to 13%, and health anxiety added a further 18.4% ($R^2 = 0.317$). Health anxiety was by far the strongest predictor ($B = 0.68$, $\beta = 0.48$, $p < 0.001$). AI use was associated with a 5.3-point higher CSS score under the inclusive definition, but not under the strict definition ($B = 1.49$, $p = 0.314$), so there was no reliable evidence of an independent AI effect. Elevated health anxiety was independently associated with higher cyberchondria (AOR = 3.12 per SD), searching for instant reassurance (AOR = 3.83) and younger age (AOR = 0.61); the model discriminated well (AUC 0.79 after optimism correction), although with about five events per predictor the odds ratios are exploratory (Table 9, Figure 6). All sensitivity analyses left these results unchanged ($r = 0.46–0.48$).

Table 8. Hierarchical robust linear regression of cyberchondria (CSS-12 total) – final model (n = 300)

Predictor	B	95% CI	β	p
Constant	12.04	6.39–17.68		<0.001
Age 18–19 (ref. 20–29)	−0.43	−2.77–1.91	−0.02	0.721
Age 30–39	**6.27**	**0.19–12.34**	**0.19**	**0.043**
Age ≥40	3.98	−0.86–8.81	0.16	0.107
Healthcare worker (ref. non-health employed)	1.06	−4.32–6.44	0.03	0.700
Medical student	0.41	−3.67–4.49	0.02	0.845
Non-health student	−0.73	−5.12–3.66	−0.03	0.745
Other health-science student	1.32	−3.61–6.25	0.06	0.599
Female	−0.05	−1.82–1.72	0.00	0.954
Chronic condition	−0.14	−2.42–2.15	−0.01	0.906
Answered mainly in English	0.51	−1.71–2.74	0.03	0.650
Uses AI chatbots	**5.33**	**2.34–8.32**	**0.20**	**<0.001**
Uses search engines	0.95	−1.00–2.90	0.06	0.342
Uses medical platforms	0.17	−2.10–2.44	0.01	0.885
Uses social media	1.46	−1.61–4.54	0.05	0.351
Trust in online/AI (per point)	0.72	−0.50–1.93	0.06	0.247
Motive: instant reassurance	−1.11	−2.91–0.69	−0.07	0.229
Motive: self-diagnosis	0.99	−1.01–3.00	0.05	0.332
Motive: cost/logistics	0.44	−1.50–2.39	0.02	0.656
Motive: need for a doctor?	0.09	−1.71–1.90	0.01	0.919
Health anxiety (SHAI, per point)	**0.68**	**0.53–0.84**	**0.48**	**<0.001**

Note. Block 1 (socio-demographics) $R^2 = 0.030$; Block 2 (+ digital use) $R^2 = 0.133$, $\Delta R^2 = 0.103$, $p = 0.0002$; Block 3 (+ SHAI) $R^2 = 0.317$, $\Delta R^2 = 0.184$, $p < 0.001$; adjusted $R^2 = 0.268$; $F = 8.03$, $p < 0.001$. HC3 robust standard errors. β: standardised coefficient.

Table 9. Multivariable logistic regression of elevated health anxiety (SHAI ≥ 18; 57 events, n = 300)

Predictor	AOR	95% CI	p
Age group (per step)	**0.61**	**0.38–0.96**	**0.034**
Female	1.23	0.63–2.40	0.551
Health-related background	1.05	0.47–2.35	0.909
Chronic condition	1.51	0.64–3.53	0.345
Answered mainly in English	0.94	0.45–1.97	0.871
Uses AI chatbots	0.61	0.22–1.69	0.346
Uses search engines	1.43	0.69–2.96	0.330
Trust in online/AI (per point)	0.74	0.46–1.21	0.232
Motive: instant reassurance	**3.83**	**1.75–8.42**	**<0.001**
Motive: self-diagnosis	0.93	0.42–2.09	0.867
Cyberchondria (per 1 SD)	**3.12**	**2.09–4.65**	**<0.001**

Note. AUC = 0.83 (apparent; optimism-corrected 0.79); McFadden pseudo-$R^2 = 0.22$; likelihood-ratio $p < 0.001$; robust (HC1) standard errors.

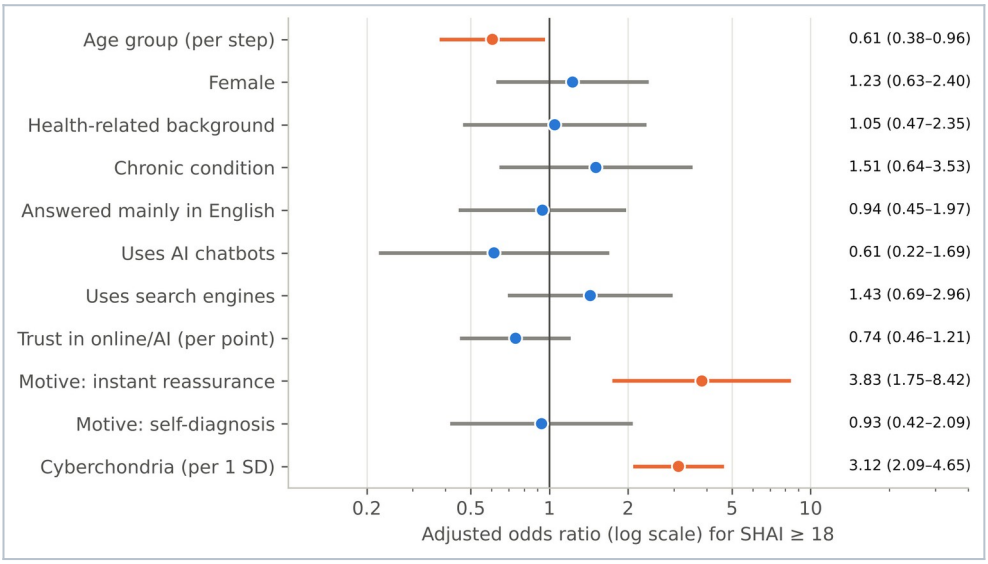


Figure 6. *Adjusted odds ratios for elevated health anxiety (SHAI ≥ 18); orange: p < 0.05.*

3.6 Behaviour and pathways

With both scores in the same models (Table 10), behaviour was related more to cyberchondria than to health anxiety: higher CSS went with urgent visits (AOR = 1.51), fear-driven delay (AOR = 1.59), less ignoring of symptoms (AOR = 0.65) and a more strained relationship (ordinal OR = 1.34). Only one of 40 tests survived FDR correction: participants with higher cyberchondria were less likely to report never telling their physician about their searches, that is, more likely to disclose them (AOR = 0.57, q = 0.012). Higher health anxiety pointed towards concealment (AOR = 1.47, q = 0.098; exploratory).

Table 10. Adjusted associations of health anxiety and cyberchondria with pre-consultation behaviours and the doctor–patient relationship

Outcome	Events	SHAI AOR (95% CI)	p	CSS AOR (95% CI)	p
Urgent visit/ER	66	0.91 (0.66–1.25)	0.551	1.51 (1.09–2.10)	0.014
Multiple doctors	76	1.06 (0.79–1.43)	0.688	1.20 (0.90–1.60)	0.208
Delay/skip care (reassured)	104	0.89 (0.66–1.19)	0.417	0.83 (0.63–1.10)	0.190
Delay care (terrified)	26	1.49 (0.94–2.36)	0.087	1.59 (1.09–2.30)	0.015
Self-medication	64	0.81 (0.56–1.16)	0.245	1.00 (0.71–1.41)	0.994
Ignore symptom	90	1.16 (0.85–1.58)	0.359	0.65 (0.48–0.89)	0.007
Never disclose to physician	95	1.47 (1.08–1.98)	0.013	0.57 (0.43–0.78)	<0.001
Negative effect on doctor relationship	104	1.03 (0.78–1.38)	0.824	1.35 (1.00–1.83)	0.049

Note. AOR per 1 SD of each score, from logistic models containing both scores plus trust, AI use, age, gender, health background, chronic disease and language. FDR-adjusted q-values are available from the authors.

In exploratory mediation models (Table 11, Figure 7), excessive searching statistically accounted for part of the link between health anxiety and post-search distress (indirect = 0.16, 95% CI 0.11–0.23), and distress for almost all of the link with reassurance-seeking consultations (direct c′ = 0.01). The reverse ordering was equally significant (indirect = 0.15), so direction cannot be established. No moderator altered the association after FDR correction, and participants with elevated health anxiety scored higher on all four dimensions (Figure 8).

Table 11. Exploratory mediation models decomposing the association between health anxiety (HA) and the dimensions of cyberchondria (standardised coefficients, n = 300)

Model	a	b	Total (c)	Direct (c′)	Indirect (95% boot CI)
A. HA → excessive searching → post-search distress	0.46	0.35	0.46	0.30	0.16 (0.11–0.23)
B. HA → excessive searching → compulsion	0.46	0.27	0.28	0.15	0.13 (0.07–0.20)
C. HA → post-search distress → reassurance seeking	0.46	0.48	0.22	0.01 (ns)	0.22 (0.16–0.29)
D. Excessive searching → distress → HA (reverse)	0.49	0.30	0.47	0.32	0.15 (0.09–0.22)
E. Serial: HA → excess → distress → reassurance				−0.04 (ns)	0.07 (0.04–0.11)

Note. a: predictor → mediator; b: mediator → outcome adjusted for the predictor; indirect effect = a × b = c − c′. All models adjusted for age, gender, health background, chronic disease, trust and AI use; 5000 bootstrap resamples. All a, b and total paths p < 0.001. Moderation of the SHAI–CSS association (interaction terms): trust in online/ai data p = 0.34; ai chatbot use p = 0.47; health-related background p = 0.66; female gender p = 0.98; chronic condition p = 0.75; english-language responder p = 0.29; age group (ordinal) p = 0.07 (all q > 0.40).

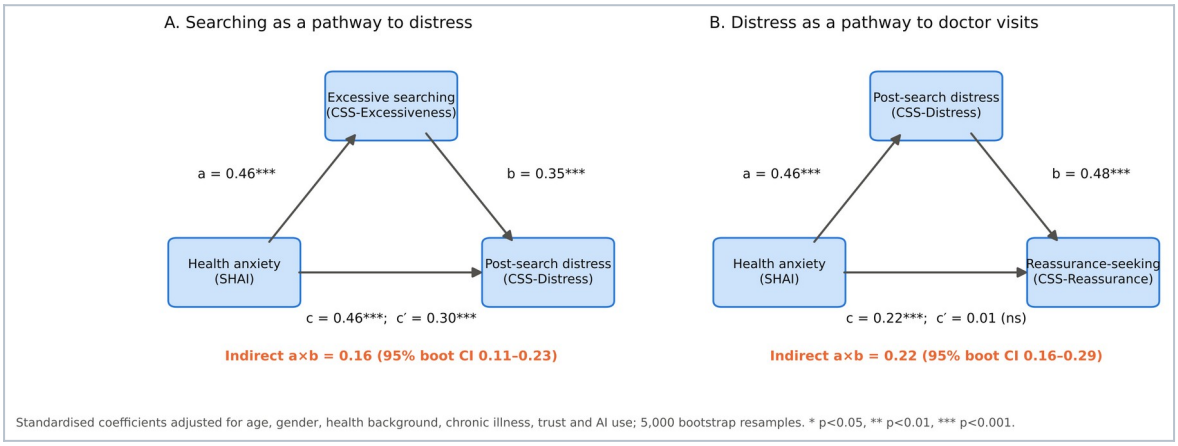


Figure 7. *Mediation models: (A) excessive searching between health anxiety and post-search distress; (B) post-search distress between health anxiety and reassurance seeking.*

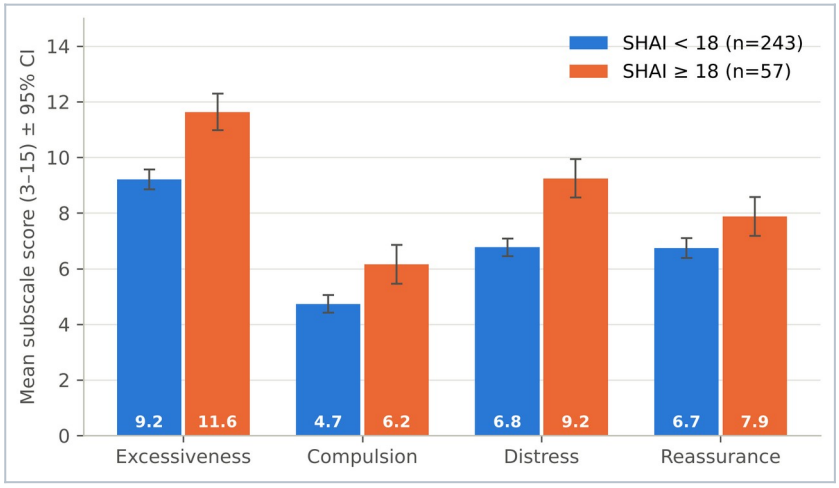


Figure 8. *Cyberchondria dimensions by health-anxiety status (mean ± 95% CI).*

4. Discussion

4.1 Health anxiety and cyberchondria

To our knowledge, this is the first study to use the CSS-12 in Baghdad and one of the first to examine AI chatbots alongside search engines before a medical visit. Health anxiety and cyberchondria were moderately and robustly associated (r = 0.47), supporting H1 and H2, although the link was weaker than the meta-analytic estimates of 0.62–0.63 [12,13]. Corrected for the reliability of both scales it would be about 0.56. Low and narrowly spread health anxiety (mean SHAI 12.3), recruitment of confident Internet users and the short CSS-12 may also have weakened it. The association was

concentrated in excessiveness and distress, and at item level in fear and catastrophic interpretation rather than bodily vigilance, as the cognitive model predicts [5]. The 19% with elevated health anxiety is a screening estimate that varied from 7% to 33% across plausible cut-offs, but it suggests that a substantial minority of digitally active adults in Baghdad report elevated health worry, in keeping with pandemic data from Iraq [7].

4.2 Do health anxiety and cyberchondria fuel each other?

The mediation results fit the reinforcing cycle proposed by the cognitive model and the Dutch longitudinal study [14]: more health anxiety goes with more searching, more searching with more distress, and distress with reassurance seeking. They must not be over-interpreted. Cross-sectional mediation can be seriously biased for longitudinal processes [35], a significant indirect effect cannot identify the causal order [36], the reverse model was equally significant, and the mediators and outcomes are subscales of one instrument [37]. The data are compatible with a cycle but cannot demonstrate it; H3 is supported statistically, not causally.

4.3 AI chatbots

AI use here (78–88.7%) was far higher than in a 2024 US survey (about one in five) [16] or a 2025 survey (21%, against 98% for search engines) [38], partly because the sample was young. The higher cyberchondria of AI users vanished under the strict definition and rested on a small, older comparison group, and AI use did not change the strength of the anxiety–cyberchondria link: an absence of evidence, not evidence of no effect. Higher trust in online/AI information went with fewer multiple-doctor consultations, suggesting that trusted AI answers may replace some "doctor shopping", but also offering a route for false reassurance; a non-significant trend towards

reassured delay (AOR = 1.33, p = 0.088) echoes regional data in which trust predicted delayed consultation [20].

4.4 The consultation and medical students

Delaying care because of reassurance (34.7%) may save visits for minor symptoms but risks late diagnosis for serious ones, and it was unrelated to either score, so it is not limited to anxious users. Almost a third never told their doctor about their searches. Higher cyberchondria robustly went with more disclosure, whereas health anxiety tended towards concealment (exploratory), perhaps from fear of being labelled the "worried well"; information discussed openly strengthens the relationship, whereas hidden information undermines it [17]. H4 is only partly supported. Medical students' higher health anxiety disappeared after adjustment for age, and their cyberchondria equalled that of others; a Palestinian study using the same SHAI threshold even found elevated health anxiety least common among medical students [39]. These results agree with Arab-region data from Jeddah, Saudi primary care, Lebanon and 1435 Egyptian medical students, of whom 85.8% showed moderate cyberchondria [9,18,19,40].

4.5 Strengths and limitations

Strengths include two established, reliable scales, formal Arabic–English invariance testing, separate assessment of AI chatbots, complete data, correction for multiple testing, robust and bootstrap methods, and sensitivity analyses. The main limitations are:

- **Design:** cross-sectional data cannot establish direction; mediation describes statistical pathways only.
- **Sample:** an online convenience sample, young and with many medical students (44.0%), so percentages cannot be generalised to Baghdad; residence was not verified by a question.

- **Measurement:** Arabic wording without formal back-translation (although the CSS-12 was fully invariant); one SHAI item with reordered options; a poorly fitting one-factor SHAI-14 model; CSS item 3 mentions only search engines and may miss repeated chatbot use; behaviour, trust and disclosure measured with single, non-validated items; all data self-reported.
- **Analysis:** many exploratory tests despite FDR correction; small subgroups (34 AI non-users); about five events per predictor in the logistic model; the AI result depends on how 32 respondents were classified.

5. Conclusions and Recommendations

5.1 Conclusions

1. AI chatbots, mainly ChatGPT, are a leading source of health information before consulting a doctor among digitally active adults in Baghdad (78.0% ticked; 88.7% including those who named one).
2. About one in five (19%) had elevated health anxiety on a screening threshold; younger people and reassurance seekers were most affected.
3. Health anxiety and cyberchondria are moderately and robustly associated ($r = 0.47$), compatible with a reinforcing cycle that this design cannot prove.
4. There is no reliable evidence that AI use adds to cyberchondria beyond health anxiety.
5. Searching shapes the consultation: about a third delay care when reassured and about a third never disclose their searches; higher cyberchondria goes with more disclosure.

5.2 Recommendations

- **Clinicians:** ask every patient, without judgement, what they have searched or asked an AI; treat repeated searching and reassurance seeking as possible signs of health anxiety; teach digital and AI health literacy in medical school.
- **Ministry of Health:** the Ministry may consider Arabic digital and AI health-literacy campaigns that warn about false reassurance, and reliable Arabic triage sources.
- **AI developers:** developers may consider anxiety-aware health answers that avoid escalating rare diagnoses and encourage appropriate consultation.
- **Research:** longitudinal studies, probability samples and formally adapted Arabic scales; intervention trials are needed, as ClinicalTrials.gov lists only nine cyberchondria studies and none in Iraq [41].

6. References

1. White RW, Horvitz E. Cyberchondria: studies of the escalation of medical concerns in Web search. ACM Trans Inf Syst. 2009;27(4):1-37. doi:10.1145/1629096.1629101
2. Starcevic V, Berle D. Cyberchondria: towards a better understanding of excessive health-related Internet use. Expert Rev Neurother. 2013;13(2):205-13. doi:10.1586/ern.12.162
3. Kemp S. Digital 2025: Iraq [Internet]. DataReportal; 2025 [cited 2026 Oct 15]. Available from: https://datareportal.com/reports/digital-2025-iraq
4. American Psychiatric Association. Diagnostic and statistical manual of mental disorders. 5th ed., text revision (DSM-5-TR). Washington (DC): American Psychiatric Association Publishing; 2022.
5. Salkovskis PM, Warwick HMC. Morbid preoccupations, health anxiety and reassurance: a cognitive-behavioural approach to hypochondriasis. Behav Res Ther. 1986;24(5):597-602. doi:10.1016/0005-7967(86)90041-0
6. Tyrer P. Recent advances in the understanding and treatment of health anxiety. Curr Psychiatry Rep. 2018;20(7):49. doi:10.1007/s11920-018-0912-0
7. Karim SK, Taha PH, Amin NMM, Ahmed HS, Yousif MK, Hallumy AM. COVID-19-related anxiety disorder in Iraq during the pandemic: an online cross-sectional study. Middle East Curr Psychiatry. 2020;27(1):55. doi:10.1186/s43045-020-00067-4

8. McElroy E, Kearney M, Touhey J, Evans J, Cooke Y, Shevlin M. The CSS-12: development and validation of a short-form version of the cyberchondria severity scale. Cyberpsychol Behav Soc Netw. 2019;22(5):330-5. doi:10.1089/cyber.2018.0624
9. Hallit S, Rogoza R, Abi Semaan C, Azzi V, Sawma T, Obeid S. Validation of the Arabic version of the cyberchondria severity scale 12 items (CSS-12-Ar) among a sample of Lebanese adults. BMC Psychiatry. 2023;23(1):618. doi:10.1186/s12888-023-05123-x
10. Mathes BM, Norr AM, Allan NP, Albanese BJ, Schmidt NB. Cyberchondria: overlap with health anxiety and unique relations with impairment, quality of life, and service utilization. Psychiatry Res. 2018;261:204-11. doi:10.1016/j.psychres.2018.01.002
11. Jungmann SM, Brand S, Kolb J, Witthöft M. Do Dr. Google and health apps have (comparable) side effects? An experimental study. Clin Psychol Sci. 2020;8(2):306-17. doi:10.1177/2167702619894904
12. McMullan RD, Berle D, Arnáez S, Starcevic V. The relationships between health anxiety, online health information seeking, and cyberchondria: systematic review and meta-analysis. J Affect Disord. 2019;245:270-8. doi:10.1016/j.jad.2018.11.037
13. Schenkel SK, Jungmann SM, Gropalis M, Witthöft M. Conceptualizations of cyberchondria and relations to the anxiety spectrum: systematic review and meta-analysis. J Med Internet Res. 2021;23(11):e27835. doi:10.2196/27835
14. te Poel F, Baumgartner SE, Hartmann T, Tanis M. The curious case of cyberchondria: a longitudinal study on the reciprocal relationship between health anxiety and online health information seeking. J Anxiety Disord. 2016;43:32-40. doi:10.1016/j.janxdis.2016.07.009
15. Ayers JW, Poliak A, Dredze M, Leas EC, Zhu Z, Kelley JB, et al. Comparing physician and artificial intelligence chatbot responses to patient questions posted to a public social media forum. JAMA Intern Med. 2023;183(6):589-96. doi:10.1001/jamainternmed.2023.1838
16. Ayo-Ajibola O, Davis RJ, Lin ME, Riddell J, Kravitz RL. Characterizing the adoption and experiences of users of artificial intelligence-generated health information in the United States: cross-sectional questionnaire study. J Med Internet Res. 2024;26:e55138. doi:10.2196/55138
17. Luo A, Qin L, Yuan Y, Yang Z, Liu F, Huang P, et al. The effect of online health information seeking on physician-patient relationships: systematic review. J Med Internet Res. 2022;24(2):e23354. doi:10.2196/23354
18. El-Zayat A, Namnkani SA, Alshareef NA, Mustfa MM, Eminaga NS, Algarni GA. Cyberchondria and its association with smartphone addiction and electronic health literacy among a Saudi population. Saudi J Med Med Sci. 2023;11(2):162-8. doi:10.4103/sjmms.sjmms_491_22
19. Pasay-An E, Alalawi RH, Shanmugam SR, Alshammari SA, Alreshidi MS, Alkubati S, et al. "Dr. Google" consultation and health anxiety among primary healthcare

patients: the mediation effect of health literacy. PLoS One. 2025;20(10):e0325791. doi:10.1371/journal.pone.0325791

20. Dahshan A, Mahfouz Khalil MI, Abdelfatah D, Abu Ejheisheh M, Alshammari MSH, Mekdad AK, et al. Influence of online health information and artificial intelligence on decision-making among migraine patients: a multinational cross-sectional study from the MENA region. Cephalalgia. 2026;46(4):3331024261435594. doi:10.1177/03331024261435594
21. von Elm E, Altman DG, Egger M, Pocock SJ, Gøtzsche PC, Vandenbroucke JP. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. Lancet. 2007;370(9596):1453-7. doi:10.1016/S0140-6736(07)61602-X
22. Eysenbach G. Improving the quality of Web surveys: the Checklist for Reporting Results of Internet E-Surveys (CHERRIES). J Med Internet Res. 2004;6(3):e34. doi:10.2196/jmir.6.3.e34
23. Serdar CC, Cihan M, Yücel D, Serdar MA. Sample size, power and effect size revisited: simplified and practical approaches in pre-clinical, clinical and laboratory studies. Biochem Med (Zagreb). 2021;31(1):010502. doi:10.11613/BM.2021.010502
24. Salkovskis PM, Rimes KA, Warwick HMC, Clark DM. The Health Anxiety Inventory: development and validation of scales for the measurement of health anxiety and hypochondriasis. Psychol Med. 2002;32(5):843-53. doi:10.1017/S0033291702005822
25. Arnáez S, García-Soriano G, López-Santiago J, Belloch A. The Spanish validation of the Short Health Anxiety Inventory: psychometric properties and clinical utility. Int J Clin Health Psychol. 2019;19(3):251-60. doi:10.1016/j.ijchp.2019.05.003
26. Österman S, Axelsson E, Lindefors N, Hedman-Lagerlöf E, Hedman-Lagerlöf M, Kern D, et al. The 14-item short health anxiety inventory (SHAI-14) used as a screening tool: appropriate interpretation and diagnostic accuracy of the Swedish version. BMC Psychiatry. 2022;22:701. doi:10.1186/s12888-022-04367-3
27. Cruchinho P, López-Franco MD, Capelas ML, Almeida S, Bennett PM, Miranda da Silva M, et al. Translation, cross-cultural adaptation, and validation of measurement instruments: a practical guideline for novice researchers. J Multidiscip Healthc. 2024;17:2701-28. doi:10.2147/JMDH.S419714
28. Alshayea AK. Latent structure, measurement invariance, and reliability of an Arabic version of the Short Health Anxiety Inventory. J Exp Psychopathol. 2020;11(2). doi:10.1177/2043808720912629
29. Hayes AF, Coutts JJ. Use omega rather than Cronbach's alpha for estimating reliability. But… Commun Methods Meas. 2020;14(1):1-24. doi:10.1080/19312458.2020.1718629
30. Leitgöb H, Seddig D, Asparouhov T, Behr D, Davidov E, De Roover K, et al. Measurement invariance in the social sciences: historical development,

methodological challenges, state of the art, and future perspectives. Soc Sci Res. 2023;110:102805. doi:10.1016/j.ssresearch.2022.102805

31. Ćosić Pilepić A, Mohorić T, Takšić V. Multidimensional polytomous DIF detection methods: a Monte Carlo simulation study. Appl Psychol Meas. 2026;50(7):01466216261446320. doi:10.1177/01466216261446320
32. Steiger JH. Tests for comparing elements of a correlation matrix. Psychol Bull. 1980;87(2):245-51. doi:10.1037/0033-2909.87.2.245
33. Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. J R Stat Soc Series B Methodol. 1995;57(1):289-300. doi:10.1111/j.2517-6161.1995.tb02031.x
34. Hayes AF. Introduction to mediation, moderation, and conditional process analysis: a regression-based approach. 3rd ed. New York: Guilford Press; 2022.
35. O'Laughlin KD, Martin MJ, Ferrer E. Cross-sectional analysis of longitudinal mediation processes. Multivariate Behav Res. 2018;53(3):375-402. doi:10.1080/00273171.2018.1454822
36. Rohrer JM, Hünermund P, Arslan RC, Elson M. That's a lot to process! Pitfalls of popular path models. Adv Methods Pract Psychol Sci. 2022;5(2). doi:10.1177/25152459221095827
37. Podsakoff PM, Podsakoff NP, Williams LJ, Huang C, Yang J. Common method bias: it's bad, it's complex, it's widespread, and it's not easy to fix. Annu Rev Organ Psychol Organ Behav. 2024;11:17-61. doi:10.1146/annurev-orgpsych-110721-040030
38. Yun HS, Bickmore T. Online health information-seeking in the era of large language models: cross-sectional web-based survey study. J Med Internet Res. 2025;27:e68560. doi:10.2196/68560
39. Fasfoos A, Sunkrot MS, Jadallah I, Id'eas AH, Hoshieh M, Haytham T, et al. Health anxiety among medical and non-medical university students in Palestine: associations with intolerance of uncertainty, daily phone use, and field of study. BMC Psychol. 2026;14(1):1056. doi:10.1186/s40359-026-04751-8
40. El-Zoghby SM, Zaghloul NM, Tawfik AM, Elsherbiny NM, Shehata SA, Soltan EM. Cyberchondria and smartphone addiction: a correlation survey among undergraduate medical students in Egypt. J Egypt Public Health Assoc. 2024;99(1):7. doi:10.1186/s42506-024-00154-y
41. ClinicalTrials.gov [Internet]. Bethesda (MD): National Library of Medicine (US). Search results for "cyberchondria"; [cited 2026 Oct 16]. Available from: https://clinicaltrials.gov/search?cond=cyberchondria

Appendix. Study questionnaire

Administered online through Google Forms in English and Arabic (every item shown in both languages; the Arabic text is not reproduced here). All items were required except Q7 and Q9. ○ = one answer; ☐ = all that apply.

Part 1. Consent and socio-demographic data

1. I confirm that I am at least 18 years of age, have read the description of this study and voluntarily agree to participate; I can withdraw at any point without penalty.
○ *I explicitly agree and consent to participate*

2. Age group
○ *18–19 · 20–29 · 30–39 · 40–49 · 50–59 · 60+*

3. Gender
○ *Male · Female*

4. Highest level of education completed
○ *Currently a student · Postgraduate (Master's or PhD) · Bachelor's · Diploma/vocational · High school or below · Other*

5. Nature of current study or work
○ *Medical student · Student in another healthcare field · Medical doctor · Other healthcare professional · Student in a non-healthcare field · Employed in a non-healthcare field · Other*

6. Do you have a diagnosed chronic medical condition?
○ *Yes · No*

7. If yes, please specify (optional, free text).

Part 2. Digital health information seeking

8. Which tools do you use most frequently to look up symptoms online?
☐ *Search engine (e.g. Google) · AI chatbots · Dedicated medical platforms (e.g. WebMD, Mayo Clinic) · Social media (e.g. Facebook, Reddit, Instagram)*

9. If you use AI, which one most of the time?
○ *ChatGPT · Gemini · Claude · Other*

10. Primary motivation for searching medical information online
☐ *Convenience and easy access · High cost or difficulty of doctor visits · Instant reassurance about an unexplained symptom · Self-diagnosis or double-checking a prescription · To see if a symptom is serious enough to need a doctor · Other*

11. How much do you trust medical information generated by online tools/AI (1–5)?
○ *1 Not at all · 2 Slightly · 3 Moderately · 4 Very much · 5 Completely*

Part 3. Short Health Anxiety Inventory (SHAI-14): choose the statement that best describes you (scored 0–3)

12. Worrying about health
I do not worry about my health / occasionally worry / spend much of my time worrying / spend most of my time worrying

13. Body awareness*
I notice aches/pains as much as most people (of my age) / less than most people / more than most people / I am aware of aches/pains all the time

14. Bodily sensations
As a rule I am not aware of bodily sensations or changes / sometimes aware / often aware / constantly aware

15. Thoughts of illness
Resisting thoughts of illness is never a problem / most of the time I can resist them / I try but am often unable to / they are so strong I no longer try

16. Fear of serious illness
As a rule I am not afraid I have a serious illness / sometimes afraid / often afraid / always afraid

17. Images of illness
I do not have images of myself being ill / occasionally / frequently / constantly

18. Distraction from health thoughts
No difficulty taking my mind off my health / sometimes difficult / often difficult / nothing can take my mind off it

19. Medical reassurance
Lastingly relieved if my doctor says nothing is wrong / relieved but worries sometimes return / relieved but worries always return / not relieved

20. Hearing about illness
If I hear about an illness I never think I have it / sometimes / often / always

21. Interpretation of sensations
If I have a bodily sensation I rarely wonder what it means / often wonder / always wonder / must know what it means

22. Perceived risk
I usually feel at very low / fairly low / moderate / high risk of developing a serious illness

23. Thinking you are ill
I never / sometimes / often think I have a serious illness / I usually think I am seriously ill

24. Unexplained sensations
If I notice an unexplained sensation I do not find it difficult / sometimes / often / always find it difficult to think about other things

25. Family and friends
My family/friends would say I do not worry enough / have a normal attitude / worry too much about my health / am a hypochondriac

Options for Q13 were displayed in the order shown (the original scale lists "less than" before "as much as"); scoring followed the meaning of each statement.

Part 4. Cyberchondria Severity Scale (CSS-12): ○ Never (1) · Rarely (2) · Sometimes (3) · Often (4) · Always (5)

26. If I notice an unexplained bodily sensation, I search for it on the Internet.

27. I read different web pages about the same health condition.

28. I enter the same symptoms into a search engine on more than one occasion.

29. Researching symptoms online distracts me from reading news, sports or entertainment.

30. Researching symptoms online interrupts my work (e.g. emails, documents).

31. Researching symptoms online interrupts my offline social activities.

32. I panic when I read online that a symptom I have could be a sign of a rare or serious condition.

33. I feel fine until I read about a serious medical condition online.

34. I feel more anxious or distressed after researching symptoms online.

35. Researching symptoms online leads me to consult my general practitioner.

36. I suggest to my doctor a specific diagnostic procedure I have read about online (e.g. a biopsy, a blood test).

37. Researching symptoms online leads me to consult other medical specialists.

Part 5. Pre-consultation behaviour and the doctor–patient relationship

38. After searching your symptoms online, what are your most frequent responses?
☐ Buy self-prescribed medication without a doctor · Delay or skip medical help because the information reassured me · Delay help because the information frightened or overwhelmed me · Seek an urgent appointment or the emergency room · Visit several doctors to double-check · Ignore the symptom

39. During a consultation, do you share your online or AI findings with your physician?
○ Yes, always · Sometimes, depending on how the doctor responds · No, never, for fear of being judged or dismissed

40. How has digital health information-seeking affected your relationship with and trust in healthcare providers?
○ Significantly reduced my trust · Somewhat complicated the relationship, but I still follow advice · No negative impact; it supplements my doctor's care

41. Which language did you rely on most while completing this questionnaire?
○ Mostly English · Mostly Arabic